THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE FULL SYLLABUS PAPER SET 04

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A well-tested scientific theory: 1

(a) is final and can never be questioned
(b) is just an opinion
(c) is rejected if a student disagrees
(d) may be improved or revised when new evidence appears

2. In a simple physics model, a moving car is often represented as: 1

(a) a cell diagram **(b)** a sphere with bonds
(c) a layered sphere **(d)** a single point

3. The solubility of most solids in water generally: 1

(a) does not depend on temperature **(b)** increases with temperature
(c) decreases with temperature **(d)** becomes zero on heating

4. Fog is a mixture of: 1

(a) two solids **(b)** gas in a solid
(c) solid dust in water **(d)** tiny liquid droplets in air

5. The molecular/formula unit mass of calcium carbonate, $CaCO_3$ (formula unit mass) is (H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, S = 32, Cl = 35.5, Ca = 40): 1

(a) 100 u **(b)** 102 u **(c)** 200 u **(d)** 98 u

6. The chemical formula of aluminium oxide is: 1

(a) Al_3O_2 **(b)** AlO **(c)** Al_2O_3 **(d)** Al_2O

7. The weighted average atomic mass of chlorine (35 u : 37 u = 3 : 1) is: 1

(a) 35 u **(b)** 36 u **(c)** 37 u **(d)** 35.5 u

8. The velocity of a car changes from 15 m s^{-1} to 0 m s^{-1} in 5 s in a straight line. Its average acceleration is: 1

(a) 4 m s^{-2} **(b)** −3 m s^{-2} **(c)** 3 m s^{-2} **(d)** −6 m s^{-2}

9. A body with initial velocity 0 m s^{-1} moves with constant acceleration 3 m s^{-2} for 6 s. Its final velocity is: 1

(a) 3 m s^{-1} **(b)** 36 m s^{-1} **(c)** 18 m s^{-1} **(d)** 9 m s^{-1}

10. A passenger standing in a bus falls backwards when the bus suddenly starts because of: 1

(a) inertia of motion **(b)** weight
(c) inertia of rest **(d)** friction on the bus

11. The SI unit of force is: 1

(a) watt **(b)** joule **(c)** pascal **(d)** newton

12. Machines make work easier by changing: 1

(a) the energy created **(b)** the law of conservation

(c) the magnitude or direction of the force **(d)** the total work needed

13. Sound is produced by: 1

(a) magnetic fields **(b)** vibrating objects

(c) light waves **(d)** steady objects

14. Gametes (sex cells) are formed by: 1

(a) binary fission only **(b)** osmosis

(c) meiosis **(d)** mitosis

15. When placed in plain water, a potato piece: 1

(a) decreases in weight **(b)** increases in weight

(c) stays unchanged **(d)** burns

16. The cartilage discs between the vertebrae: 1

(a) connect bones to muscles **(b)** produce blood

(c) store calcium **(d)** act as cushions and give flexibility

17. Wheat and maize are: 1

(a) gymnosperms **(b)** monocots **(c)** dicots **(d)** ferns

18. The organ where the foetus develops is the: 1

(a) ovary **(b)** vagina **(c)** oviduct **(d)** uterus

19. Transfer of pollen from an anther to a stigma is called: 1

(a) germination **(b)** pollination **(c)** budding **(d)** fertilisation

20. Light travels at 3×10^8 m s^{-1}. In 1 s it travels: 1

(a) 4×10^8 m **(b)** $3/2 \times 10^8$ m **(c)** 6×10^8 m **(d)** 3×10^8 m

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Why does mathematics help in science? 2

22. Write the formulae of 2

(a) calcium chloride

(b) aluminium oxide.

23. A ball falls from rest for 2 s. Find its velocity (g = 10 m s^{-2}). 2

24. Find the net force when 10 N and 6 N act on a block in opposite directions. 2

25. What is a prokaryotic cell? Give an example. 2

26. What is a joint? Name two types. 2

27. What is ovulation? What happens if the egg is not fertilised? 2

28. What are CFCs? What did the Montreal Protocol do? 2

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SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. A mixture contains sand, salt and iron filings. Suggest the order of steps to separate all three. 3

30. An atom has 18 neutrons and mass number 35. Find its atomic number, name and configuration. 3

31. Why are roads on hills made to wind around? 3

32. What is timbre? Why do a sitar and a flute sound different at the same pitch and loudness? 3

33. Why do dancers need flexible joints? Name two joints used in folk dance. 3

34. Why do we need scientific names? State two rules of writing them. 3

35. Explain the formation of a water molecule by sharing of electrons. 3

OR

Why is hydrogen written first without a prefix in H_2S? Name H_2O and NH_3 using common names. 3

36. Describe the water cycle and its link with all four spheres. 3

OR

Explain the nitrogen cycle in short. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** A student compares sodium chloride and sugar. 5

(i) Which conducts electricity in aqueous solution?
(ii) Why?
(iii) Compare their melting points.
(iv) Classify them as ionic or covalent.

OR

(i) Compare ionic and covalent compounds in four points. 5
(ii) Why do ionic compounds dissolve in water while many covalent compounds do not?

38. (i) What is friction? 5
(ii) Describe the experiment to compare friction on different surfaces.
(iii) How does friction help us to walk?

OR

(i) Distinguish balanced and unbalanced forces. 5
(ii) Two teams pull a rope with 400 N and 300 N. Find the net force and direction.
(iii) Which team wins and why?

39. (i) Explain programmed cell death. 5
(ii) How do cancer cells differ from normal cells?
(iii) Why is cell division regulated?

OR

(i) Compare prokaryotic and eukaryotic cells in four points. 5
(ii) Give one example of each.
(iii) Where is the DNA of a bacterium found?

40. (i) Why is responsible behaviour important in adolescence? 5

(ii) Name two STIs and prevention.

(iii) Describe three contraceptive methods.

OR

(i) Explain the menstrual cycle with a diagram. 5

(ii) What happens if fertilisation takes place?

(iii) Why is the cycle of 28 days approximate?

END OF QUESTION PAPER **Total: 80 Marks**

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